50. Blocking highs disrupt the flow of low pressure systems across southern Australia

What are they?

Blocking highs are strong high pressure systems which have formed further south than usual and remain near stationary for an extended period of time. These highs essentially “block” the west to east progression of weather systems across southern Australia. Blocking highs are often, although not always, associated with a cut-off low which may form to the north of the blocking high, the two systems creating a blocking pattern. As frontal systems approach the blocking high, they slow down, weaken and tend to slip to the south of the high pressure system.

51. Cut-off lows bring enhanced rainfall to parts of southern Australia.

What are they?

Cut-off lows are low pressure systems which have broken away, or are cut-off, from the main belt of low pressure which lies to the south of Australia. They can be at any level in the atmosphere, and therefore may not show on the surface charts.

A cut-off low may develop when a low pressure system forms on an active cold front. Alternatively, they may form in an unstable easterly flow on the northern flank of a slow-moving or blocking high. This dual system is sometimes referred to as a blocking pair. East Coast Lows are a form of intense cut-off low which are visible on surface charts.

52. Easterly, or inland, troughs bring rainfall to central and inland parts of eastern Australia.

What are they?

Easterly troughs are a dominant feature of the synoptic pattern over Australia during the summer months. The trough is located on the lee side (inland side) of the Great Dividing Range, forming a boundary between the moist air near the coast and dry air inland. It extends through central Queensland and central New South Wales, sometimes extending right down into northern Victoria. It is partly formed by the intense heating of the land during the summer months, but the topography of the region also plays a role.

53. Frontal systems bring rainfall to southern Australia.

What are they?